



Civil Service
Learning

Open data driven policy making

AN2: Data and analysis

Welcome

Commissioned by



Civil Service
Learning

Delivered by



Dr David Tarrant

Kay Achenbach

The Open Data Institute



Civil Service
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Our founders



Sir Nigel Shadbolt
Chairman



Sir Tim Berners-Lee
President

Founded in 2012, the Open Data Institute (ODI) is an international, independent and not-for-profit organisation based in London, UK.



Aim

To equip policy makers with the knowledge and skills they need to effectively use open data in policy making



Introductions

Who are you?

What is your role in relation to data driven policy making?

What would you like to get out of this training?

Please pick a random time in minutes and seconds between 4 and 8 minutes, e.g. 7m23s and a colour from red, blue, yellow & green (note a colour can only be picked a max of 4 times)



Today

Introduction to open data driven policy making

Benefits, caveats and risks of using data in policy making

Data driven policy making in practice



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Data driven policy making



Session 1

Outcomes

1. Define data and open data.
2. Describe how data is used to inform policy making in different fields
3. Identify the benefits of using data in policy making
4. Assess the risks, caveats and limitations of using data in policy making



Exercise - 5 minutes in breakout rooms

Q1: What is data?

In as few words as possible, define 'data'.

You can use an example as an answer if you wish.

One twist: you cannot use the word **information** in your answer.

Q2: What is the difference between data and information?



What is data?

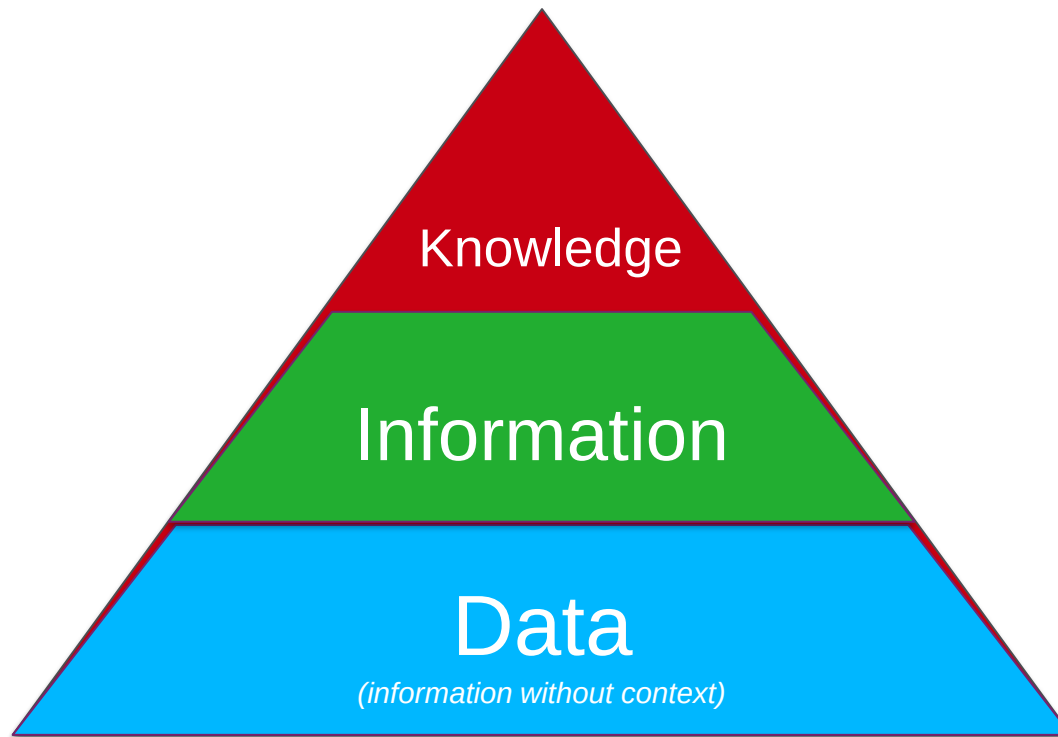
A collection of facts, information and statistics that can be analysed to develop new knowledge.

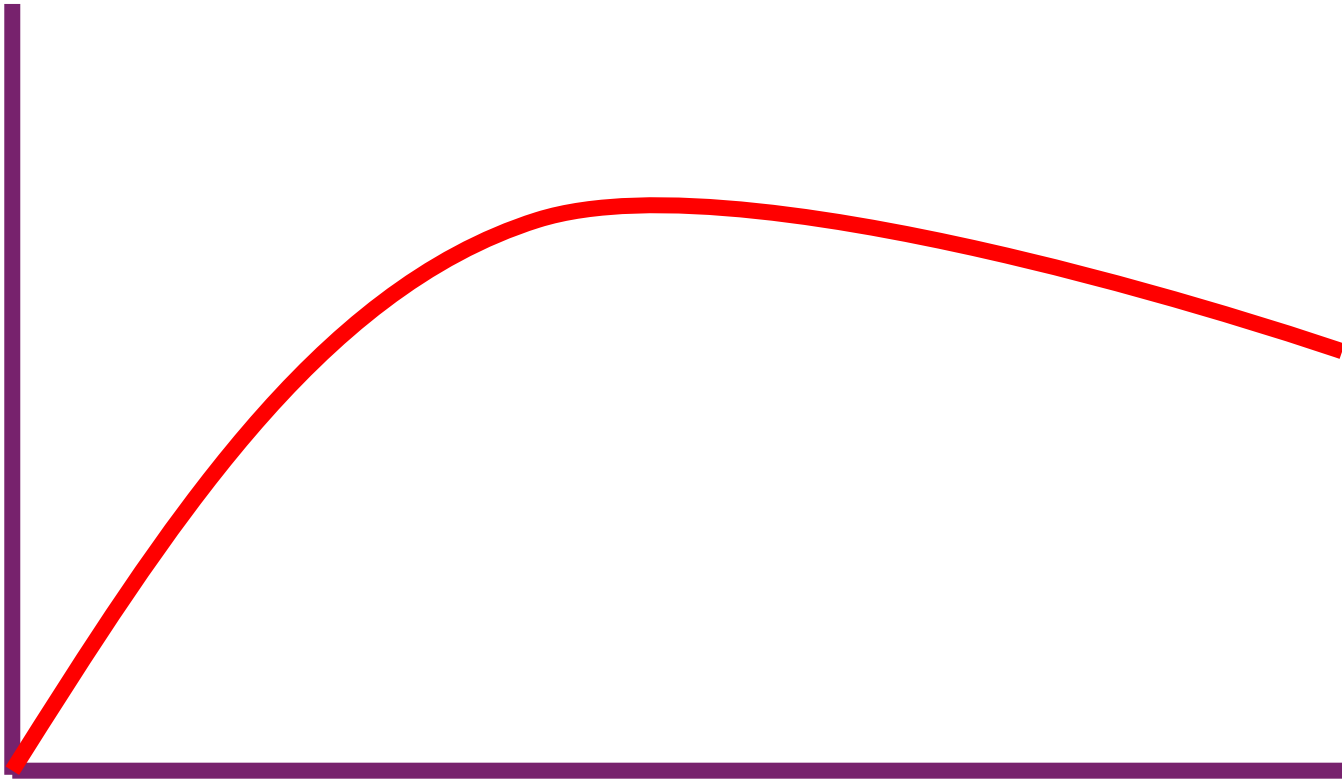
A collection of numbers assigned as values to quantitative variables and / or characters assigned as values to qualitative variables.

The lowest level of abstraction from which information and then knowledge are derived.



How is data different from information?







What did you see?

Data: What did you see?

Information: How did you interpret it?

Knowledge: Did it mean (or relate to) anything for you?



Exercise

What is open data?

In as few words as possible, what is 'open data'?



A piece of data or content is open if anyone is free to use, reuse, and redistribute it – subject only, at most, to the requirement to attribute and/or share-alike.



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Data that anyone can access, use
and share.

The Open Data Institute



Open data is data that is published in an open format, is machine readable and is published under a license that allows for free reuse.



Machine readable data?

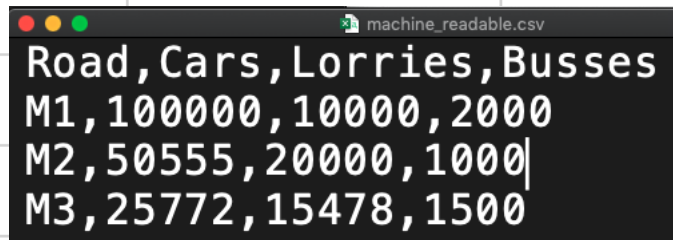
	A	B	C	D	E	F	G	H
1								
2		Traffic counts on different roads						
3						120000		
4		Road	Cars					
5		M1	100000					
6		M2	50555					
7		M3	25772					
8		Total	176327					
9								
10								
11								
12								
13								

```
''',  
,Traffic counts on different roads,,  
,  
,Road,Cars,Lorries,Busses  
,M1,100000,10000,2000  
,M2,50555,20000,1000  
,M3,25772,15478,1500  
,Total,176327,45478,4500  
,  
,  
,  
,  
,
```



Machine readable data?

	A	B	C	D
1	Road	Cars	Lorries	Busses
2	M1	100000	10000	2000
3	M2	50555	20000	1000
4	M3	25772	15478	1500
5				
6				
7				
8				



```
machine_readable.csv
Road,Cars,Lorries,Busses
M1,100000,10000,2000
M2,50555,20000,1000
M3,25772,15478,1500
```



Open data

The important points are:

- licensed openly (e.g. Open Government Licence)
- free to use, not always free to access (currently government open data must be available at no cost)
- users must be able to modify and redistribute the data

Data.gov.uk includes:

- published in open format
- machine readable



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Video

Q: What types and uses of data do Emily and David focus on?



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Case study: LIDAR

 **DATA.GOV.UK** Beta
Opening up Government

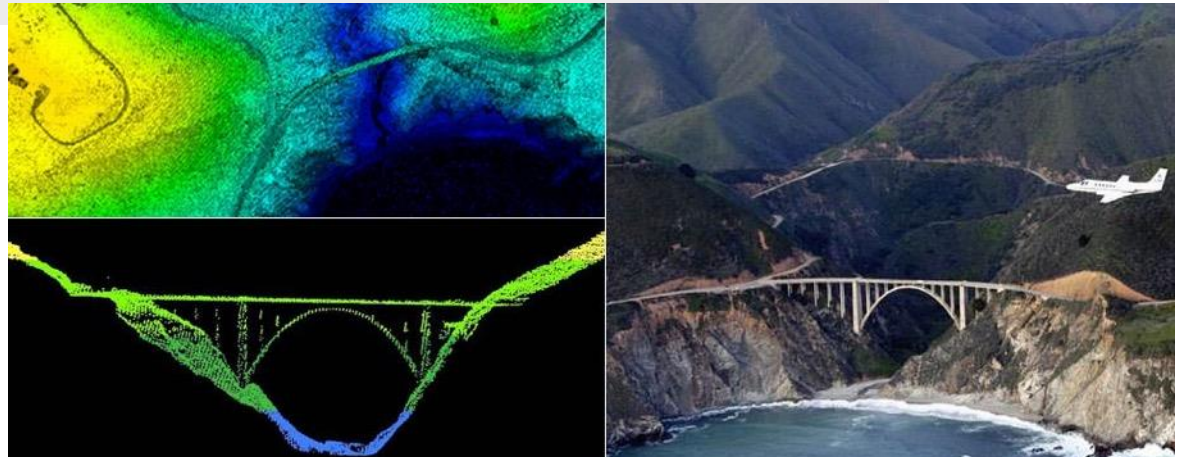
Home

[Datasets](#) [Map Search](#) [Data Requests](#) [Publishers](#) [Data API](#) [Organograms](#) [Site Analytics](#)

Home / Datasets / LIDAR Composite DTM - 50cm

LIDAR Composite DTM - 50cm

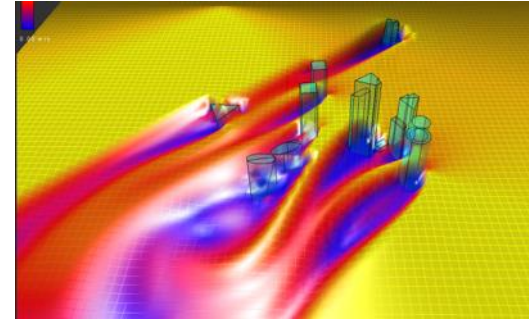
Published by Environment Agency. Licensed under **OGL** Open Government Licence.
Openness rating: ★★★★★





Impact of LIDAR data

Wind modelling



Archology



Educational games (Minecraft)





Review

What types of data do David and Emily focus on?



EMILY focusses on open data and data that is already showing benefit regardless of how big or small it is. Such as LIDAR, Transport, Environmental, Energy and other non-personal data.

DAVID focusses on Big Data and Exhaust Data that come from social interactions and from people. This brings into question lots of ethical and data protection issues and the Big Data Hubris.





Session 2

Outcomes

1. Carry out a number of different types of data analysis
 - a) Applying exploratory data analysis techniques
 - b) Evaluating shapes and trends in data
 - c) Data driven classification
2. Define big data
3. Assess the risks, caveats and limitations of using data in policy making



Exploratory data analysis

Exploratory Data Analysis refers to the critical process of performing initial investigations on data to discover patterns, spot anomalies, and to establish the shape of the data.

This helps you test hypothesis and to check assumptions with the help of summary statistics and graphical representations.



Exploratory data analysis

You each have a set of cards containing numerical values representing something. Can you work out what the dataset represents?

Data analysis stages:

Data: What do you see?

Information: What can you calculate?

Knowledge: What does it mean to you?

Clues:

The dataset is complete

The numbers are not rounded

The numbers are discreet (none are additions...etc of others)



Averages

- What problems does this dataset have when working out the average?
- How might we solve these problems?
- Is the middle value (with data points ordered) better than the mean?

Outliers

33,750.00	33,750.00
44,000.00	33,750.00
138,188.00	33,750.00
45,566.67	33,750.00
44,000.00	44,000.00
141,666.67	44,000.00
292,500.00	44,000.00
5,600,000.00	45,566.67
103,500.00	65,000.00
190,000.00	95,000.00
65,000.00	103,500.00
33,750.00	112,495.50
195,000.00	138,188.00
4,600,000.00	141,666.67
194,375.00	181,500.00
33,750.00	185,000.00
112,495.50	190,000.00
95,000.00	194,375.00
301,999.00	195,000.00
181,500.00	205,000.00
33,750.00	292,500.00
185,000.00	301,999.00
205,000.00	4,600,000.00
44,000.00	5,600,000.00

Average (mean): 518,311.64

Median: ~125k

17% of 24 = 4

17% trimmed mean: 132,314.50
=trimmean(RANGE,0.17)



5 number summary (of the trimmed dataset)

33,750	33,750	33,750	33,750	44,000
44,000	44,000	45,566	65,000	95,000
103,500	112,495	138,188	181,500	185,000
141,666	190,000	194,375	195,000	205,000
292,500	301,999	4,000,000	5,000,000	

With the cards in order, note down the following values:

1. Minimum value
2. Maximum value
3. Median
4. Value $\frac{1}{4}$ of the way through
5. Value $\frac{3}{4}$ of the way through



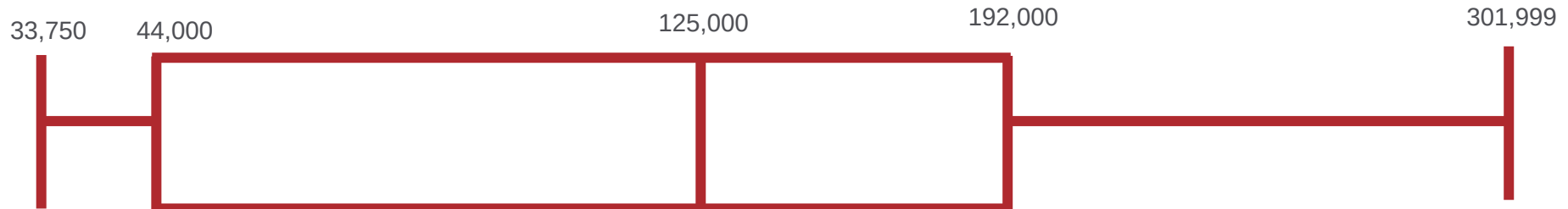


5 number summary (of the trimmed dataset)

33,750	33,750	33,750	33,750	44,000
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With the cards in order, note down the following values:

1. Minimum value
2. Maximum value
3. Median
4. Value $\frac{1}{4}$ of the way through
5. Value $\frac{3}{4}$ of the way through





The distribution of data

Divide the data into 6 evenly distributed buckets of values and stack the cards as per the diagram here.

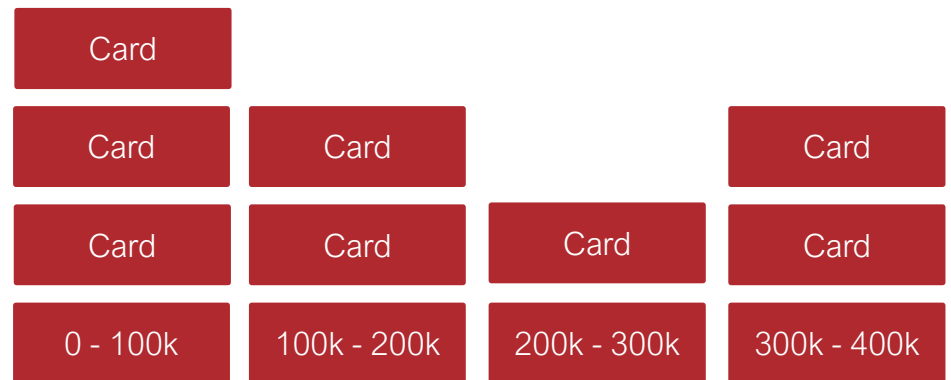
Suggested buckets:

0 – 100,000

100,000 – 200,000

200,000 - 300,000

...

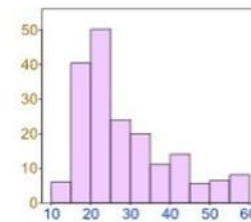




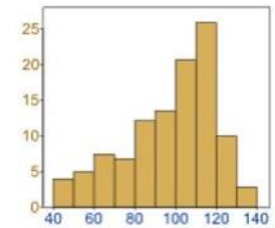
The distribution of data

Another important aspect to consider is the distribution of data. It is always a good practice to know the distribution of your data before analysing it further. Certain analyses require certain distributions.

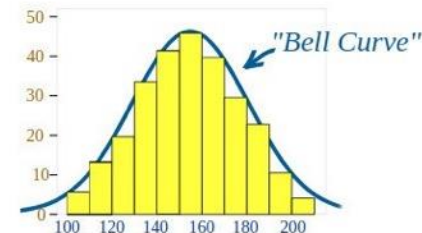
The examples here show different types of distributions of data.



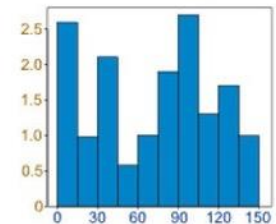
Positively skewed
distribution



Negatively skewed
distribution



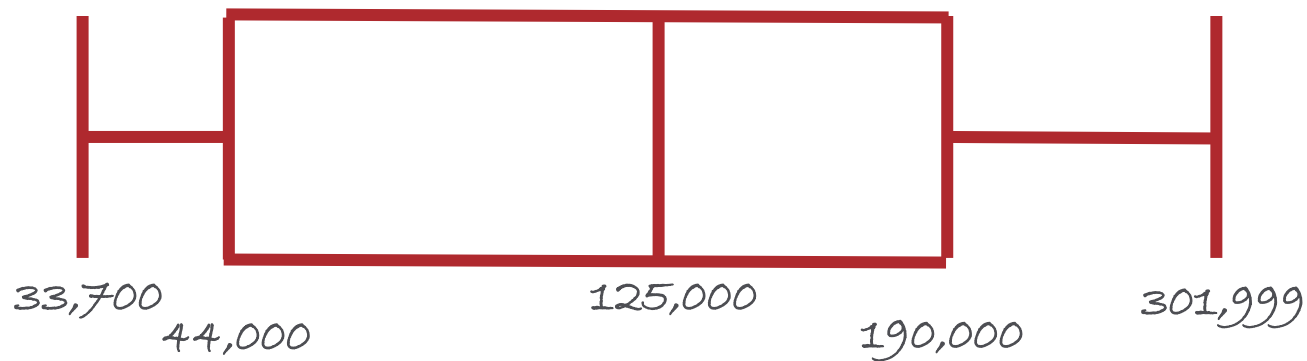
Normal distribution



Non-normal
distribution

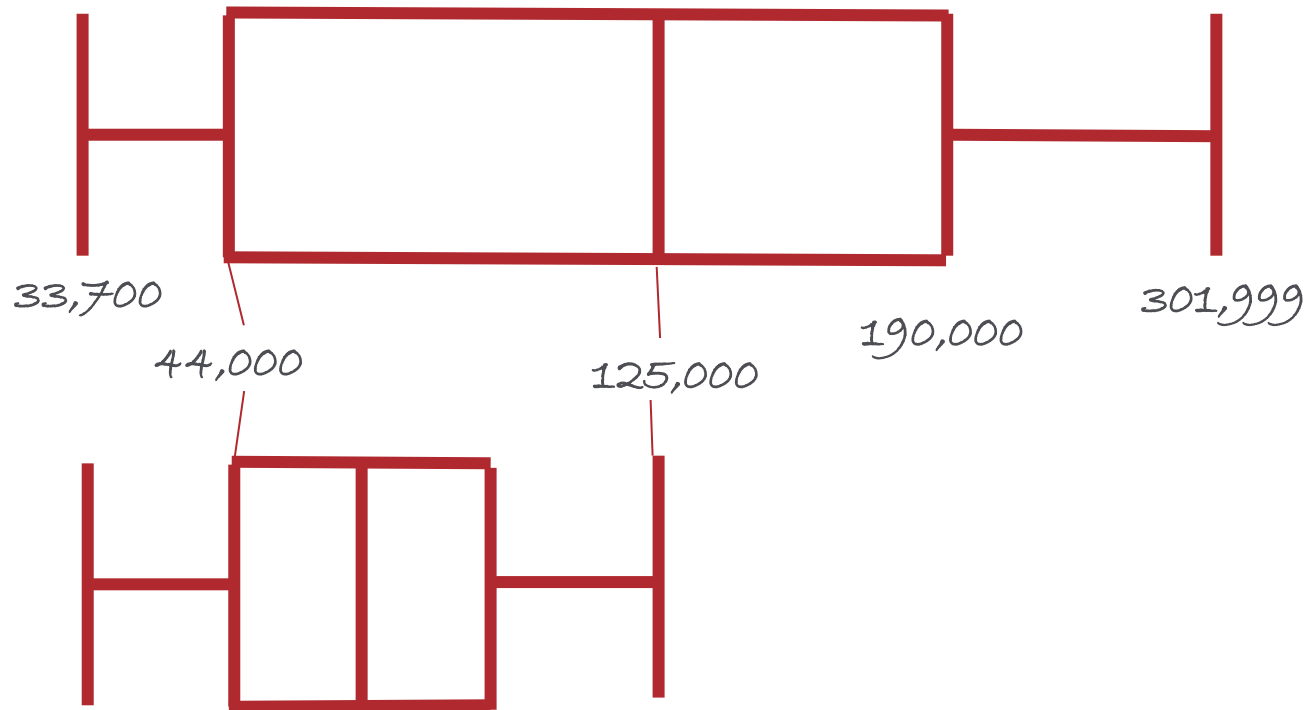


5 number summary (of the trimmed dataset)





5 number summary (of the trimmed dataset)





Five-number summary in Tableau



Connect

To a File

- Microsoft Excel
- Text file
- JSON file
- PDF file
- Spatial file
- Statistical file

To a Server

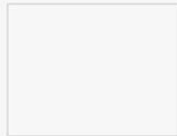
- OData
- More...



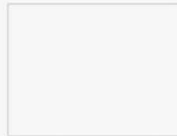
Save locally. Work with big data.
Connect to more data sources.

Upgrade Now

Open



LFB_All_Data



LFB_Trial1



London_boroughs



Greater Manche...

Open from Tableau Public



Firts 100 goals



Discover

How-to Videos

Overview

Intro to the Interface

Chart Types

More how-to videos...

Viz of the Day

THE POTTERVERSE - FAMILY TREE

Blog - Step and jump into Tableau
Public 2018.1

Sample Data Sets

Live Training

Current Status



Summary

Data, it turns out, has shape. That shape has meaning.

Exploratory Data Analysis helps you establish the shape of the data and provides key summary statistics and plots that can help test hypothesis and establish the suitability of the data for use in policy making.





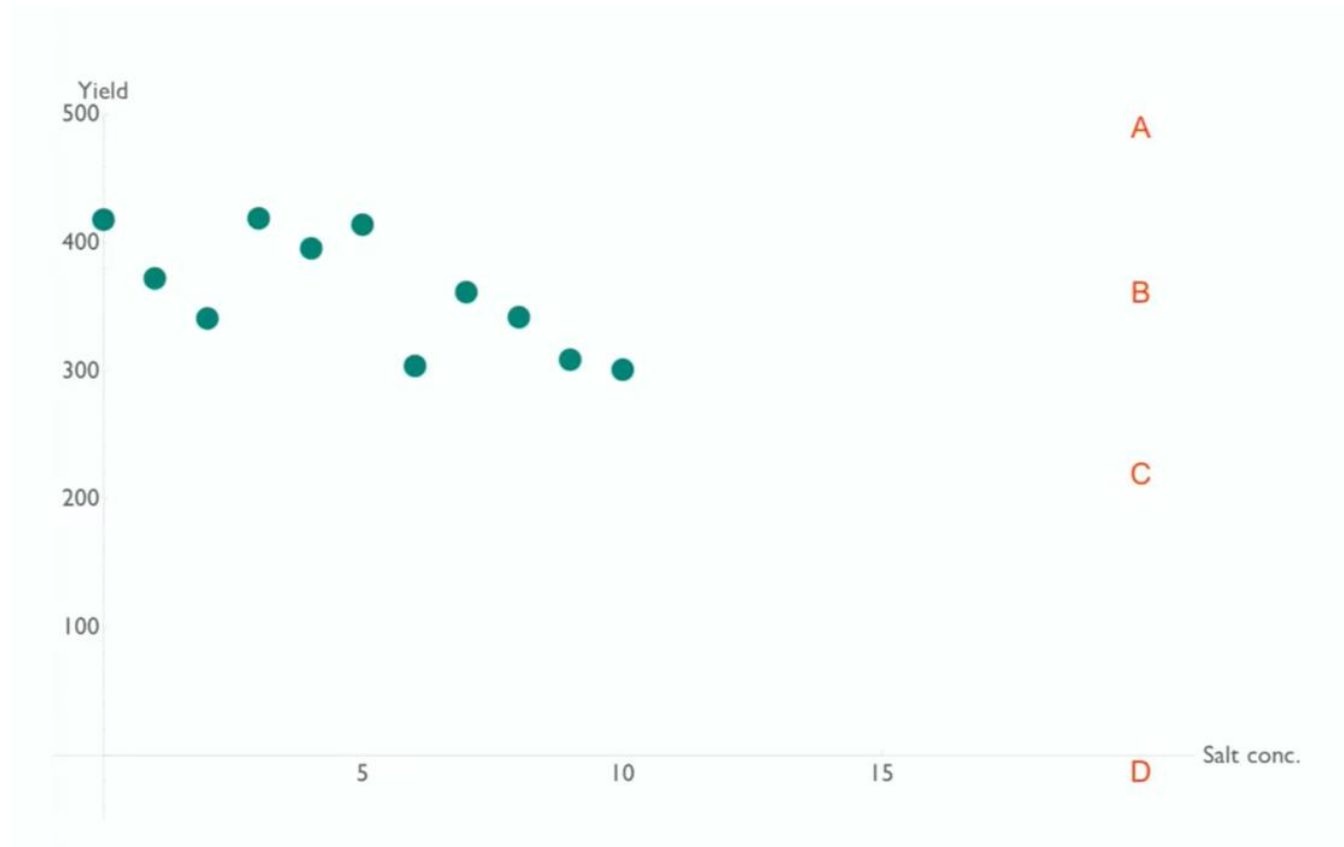
Session 2

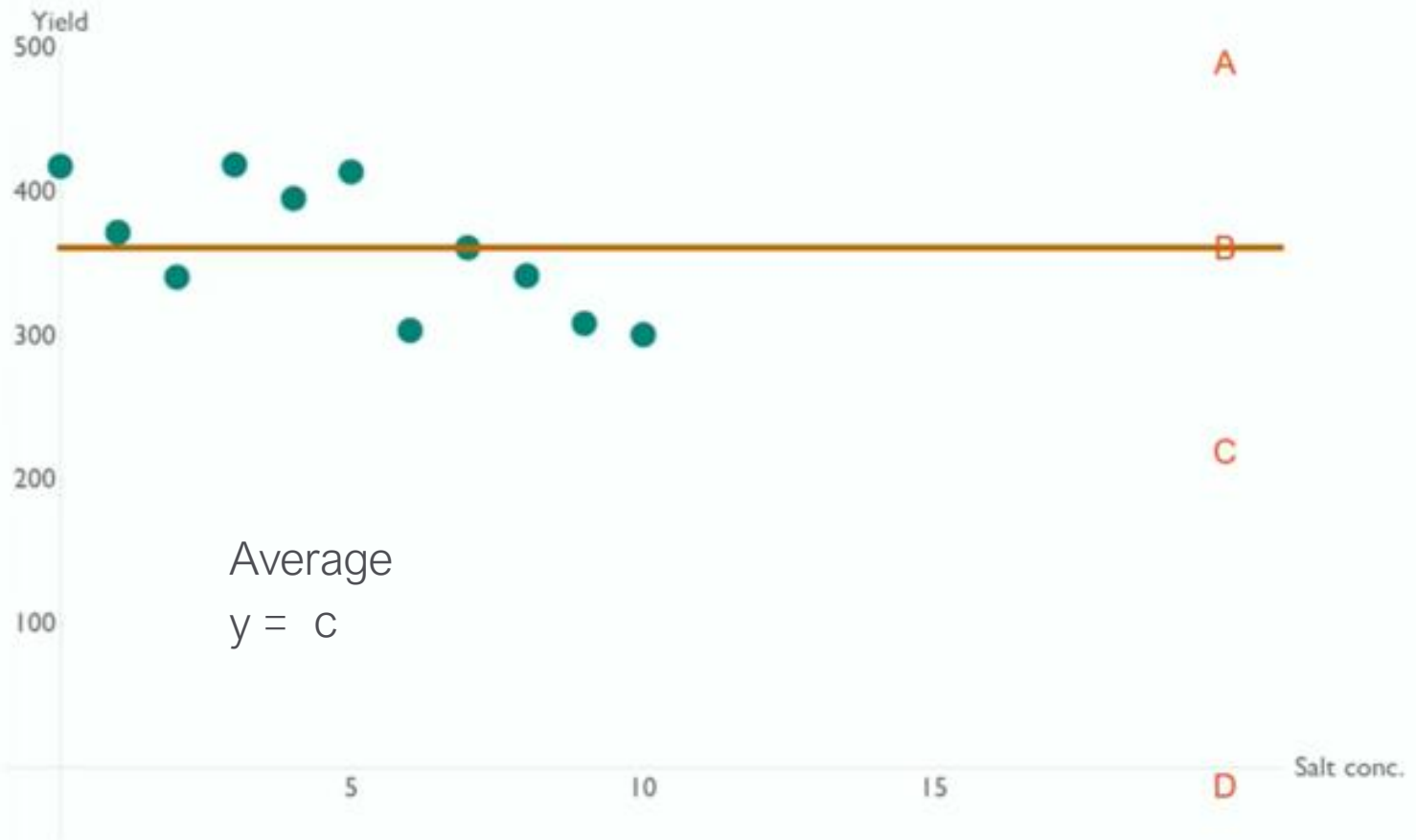
Outcomes

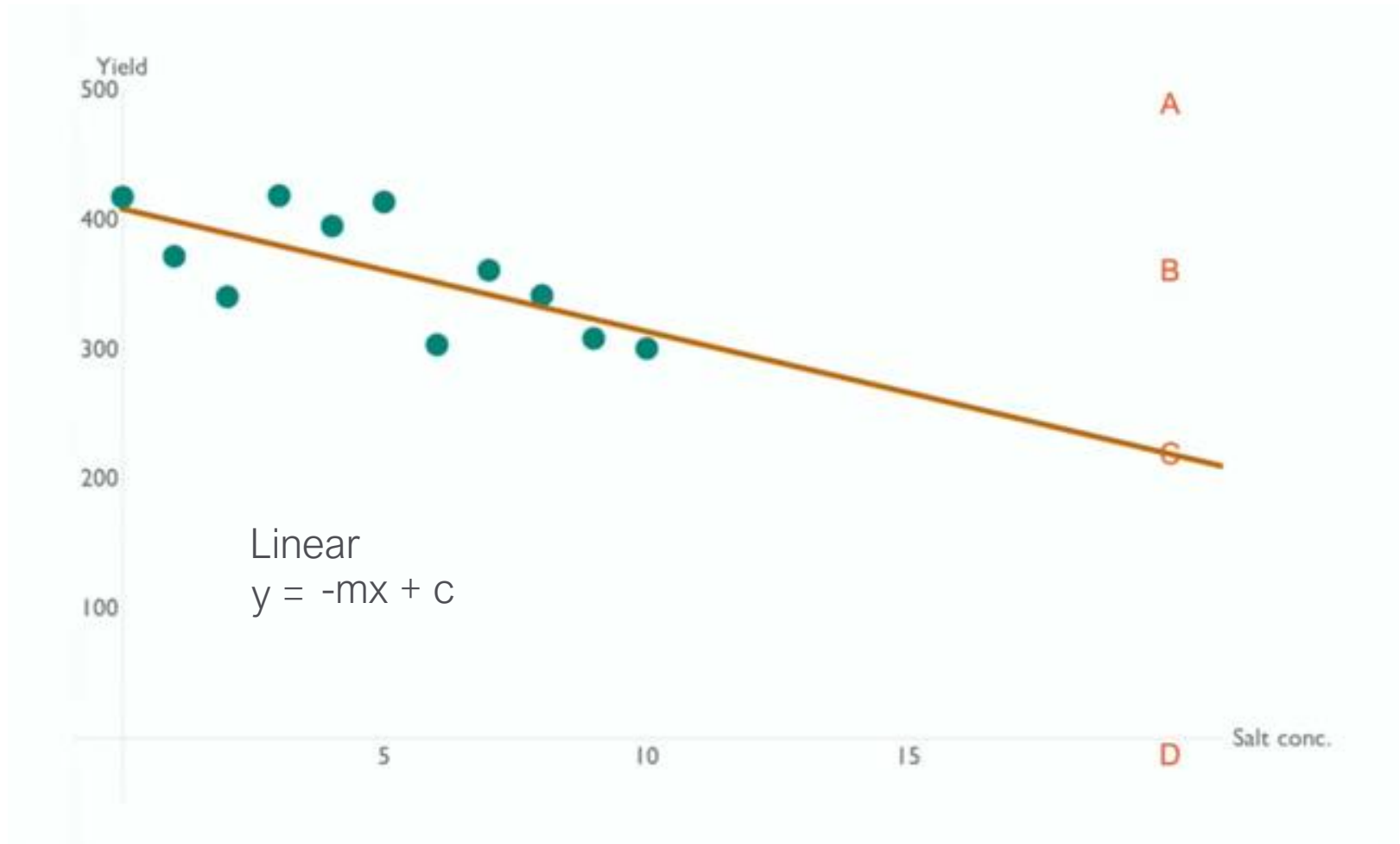
1. Carry out a number of different types of data analysis
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 - c) Data driven classification
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3. Assess the risks, caveats and limitations of using data in policy making

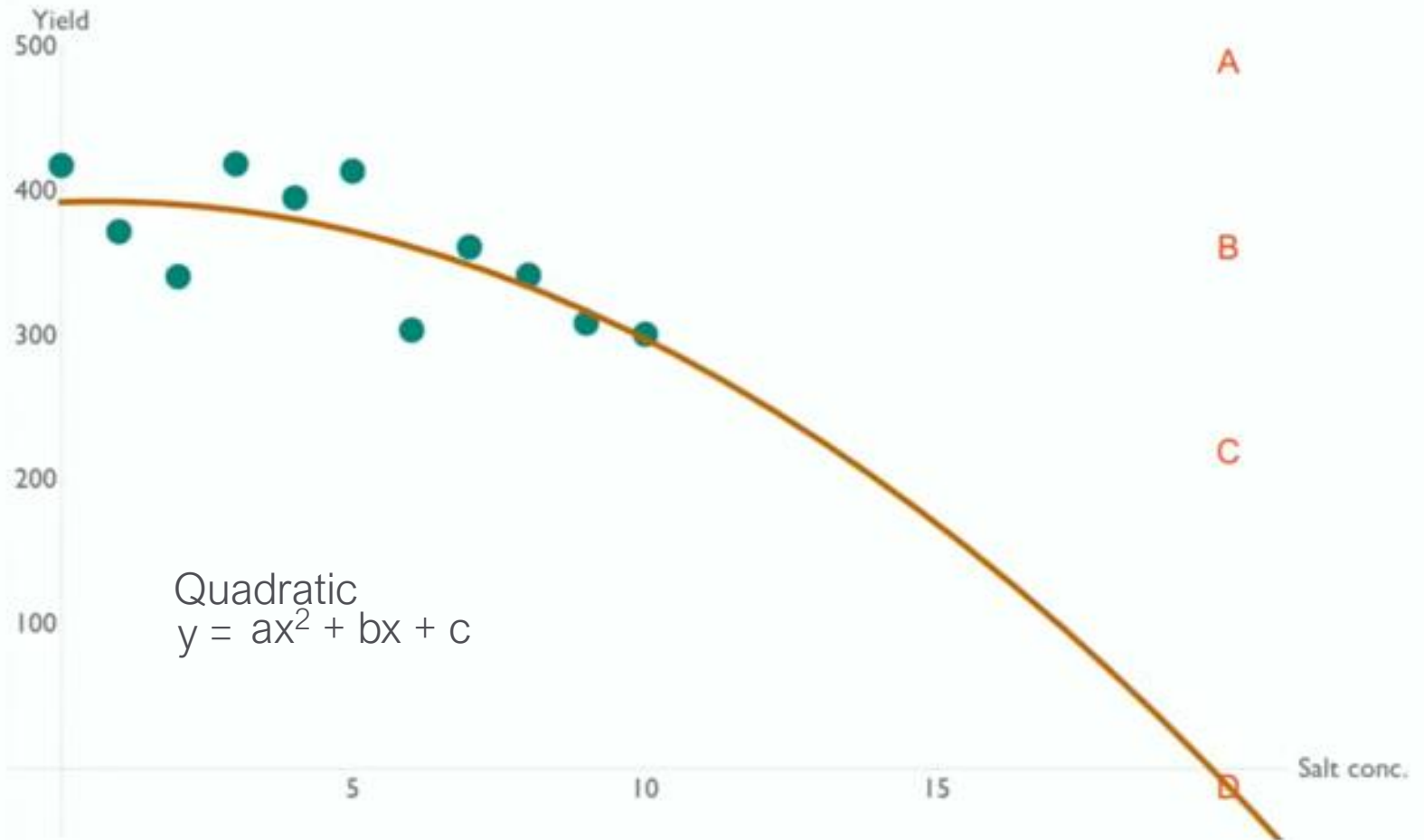


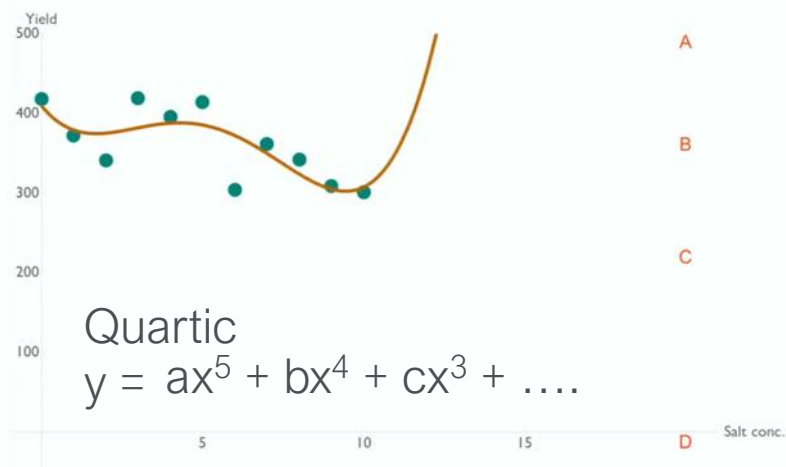
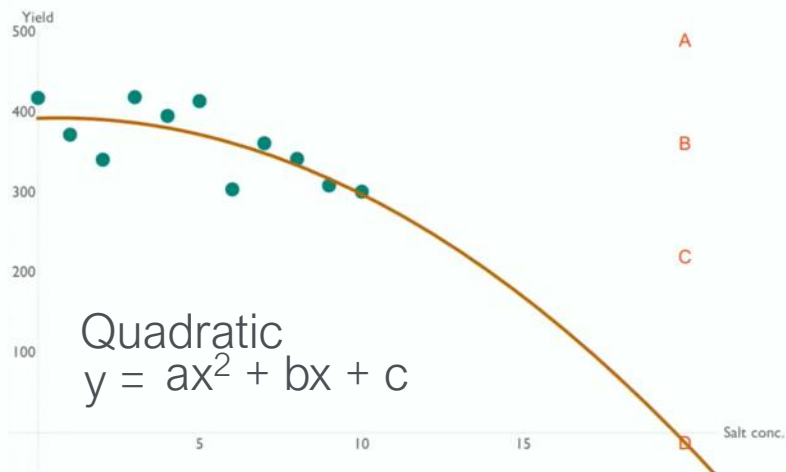
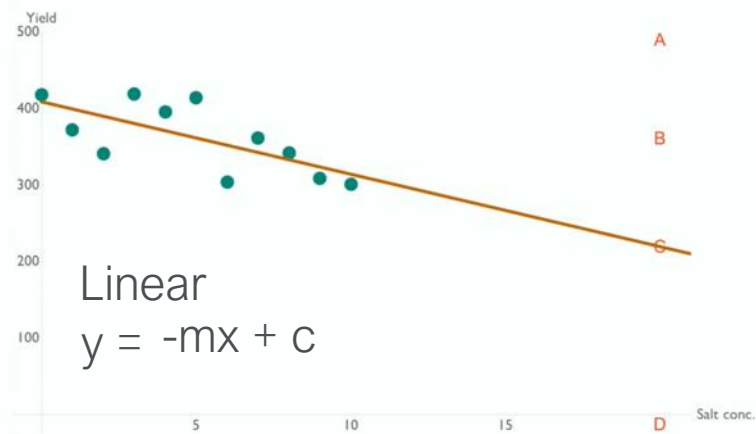
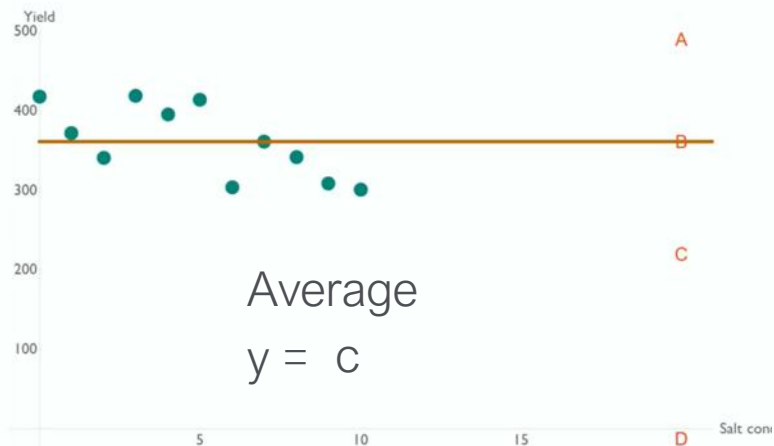
Trend analysis





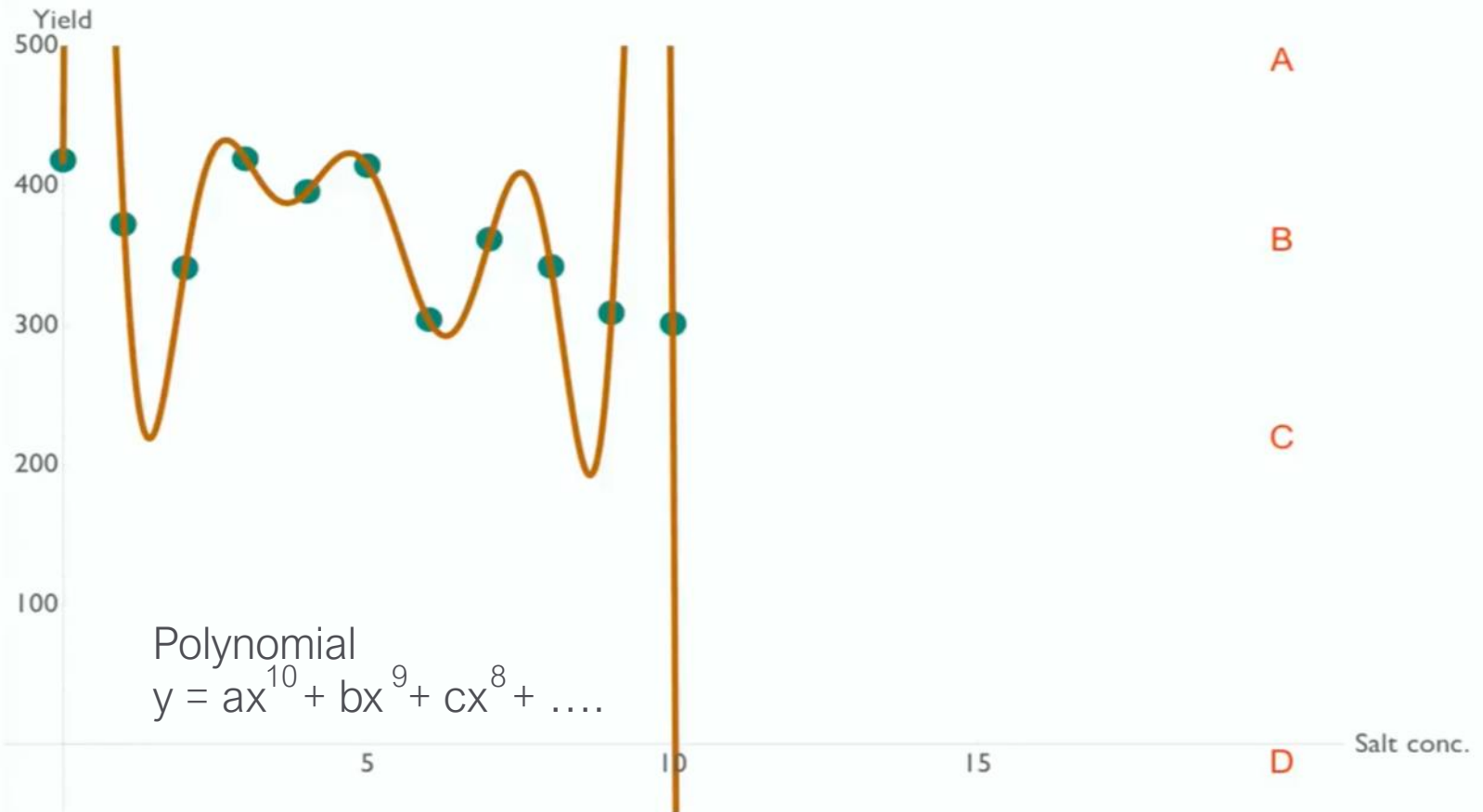








Overfitting







Session 2

Outcomes

1. Carry out a number of different types of data analysis
 - a) Applying exploratory data analysis techniques
 - b) Evaluating shapes and trends in data
 - c) **Data driven classification**
2. Define big data
3. Assess the risks, caveats and limitations of using data in policy making



Can this be done?





Classification - Exercise

Each group has a set of twenty blue cards containing data relating to properties in two cities.

The objective is to build a classification system that can categorize red cards into the two cities.

You know the city relating to the blue cards, you don't know the city related to the red cards.

This is a competitive task and there will be winners and losers.

New York

#52



Bathrooms	1
Bedrooms	1
Year built	1969
Elevation	23ft
Square Footage	600
Price	\$1,200,000
Price per sqft	\$2,000

?

#33



Bathrooms	3
Bedrooms	2
Year built	1925
Elevation	10ft
Square Footage	3,200
Price	\$2,999,000
Price per sqft	\$937



Instructions

Build a decision tree based upon the blue cards that you think can classify the red cards.

Use the MIRO boards to help you.

Please follow the guide in MIRO and on the related slide on how to build the tree and please include all labels.

There will be winners and losers to the game, however I haven't told you how to win... see if you can work it out. If you can your classification is more likely to be robust and your policy acceptable.

This normally takes 45 minutes to 1 hour.

New York

#52



Bathrooms	1
Bedrooms	1
Year built	1969
Elevation	23ft
Square Footage	600
Price	\$1,200,000
Price per sqft	\$2,000

?

#33

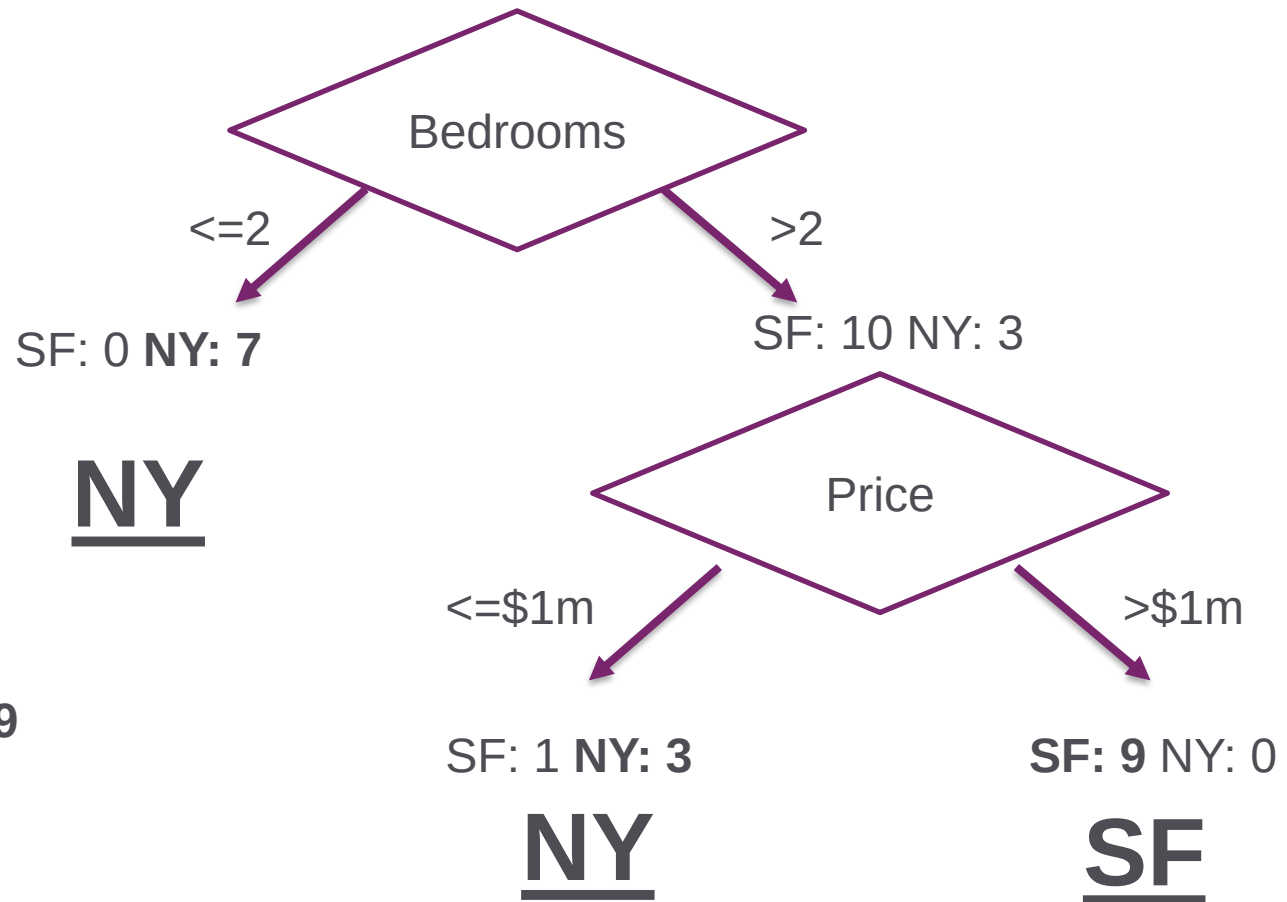


Bathrooms	3
Bedrooms	2
Year built	1925
Elevation	10ft
Square Footage	3,200
Price	\$2,999,000
Price per sqft	\$937



Example decision tree

95%
Confidence



Correct: 19
Wrong: 1



Session 3

Outcomes

1. Assess the risks, caveats and limitations of using data in policy making
 - a) Assumptions
 - b) Overfitting
 - c) Getting distracted by outliers
 - d) Making it too complicated
2. Evaluate the legal and ethical implications of using (or not using) data in policy making.



Approaches



Data first



Knowledge first



Assumption

What share of income tax paid in the UK is paid by the top 1% of earners?

♦ A: 5%

♦ B: 9%

♦ C: 14%

♦ D: 17%



Assumption

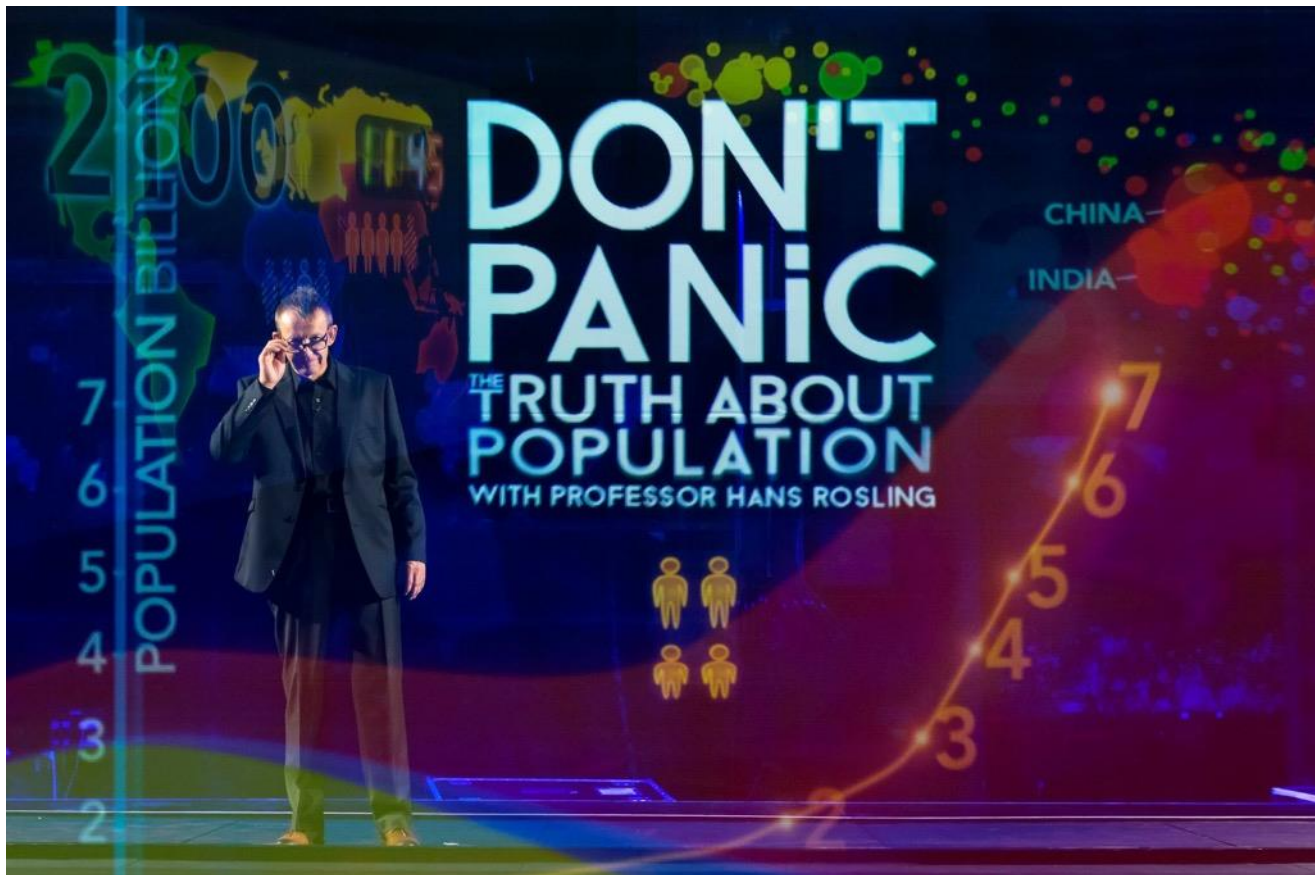
What is the average number of children per family in Bangladesh?

♦ A: 2

♦ B: 3

♦ C: 4

♦ D: 5



gapminder.org



Data analysis of the houses dataset in Tableau

Tableau Public - Book1

Data Analytics

houses

Dimensions

- Target
- Year Built**
- Zip
- Measure Names

Measures

- Bath
- Beds
- Elevation
- Index
- Price
- Price Per Sqft
- Sqft
- Latitude (generated)
- Longitude (generated)
- Number of Records
- Measure Values

Columns

Rows

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Sheet 1

Drop field here

Drop field here

Drop field here

Data Source **Sheet 1**



Options?

Grow the decision tree until...

1. every classification is perfect?
2. confidence level is above 80%?
3. it is too big to process the data in reasonable time?
4. Until the evaluation and training set have the same confidence?
5. You have used all principal components?



A visual introduction to machine learning

r2d3.us





Session 3

Outcomes

1. Assess the risks, caveats and limitations of using data in policy making
 - a) Assumptions
 - b) Overfitting
 - c) Getting distracted by outliers
 - d) Making it too complicated
2. Evaluate the legal and ethical implications of using (or not using) data in policy making.



Good or bad ideas?

1. a tool that analyses the sentiment of a user's tweets, assesses whether they are suicidal and alerts friends	4. a risk-assessment tool that uses AI to advise on prison sentences based upon criminal profile analysis
2. automatic pricing algorithm for taxi firm which responds to surges in demand	5. using energy efficiency data and winter fuel allowance data to target efficiency advice
3. using performance data to advise on how to save money in the emergency services	6. publishing genomes of 100,000 individuals for use in public health



GoCompare

ALL Details EXACTLY the same

Hi John!

Adjust cover

49 car quotes found, 4 telematics quotes included.

Sort: Annually



Legal
Assistance
+£30.99



Breakdown
Cover
+£43.99



Personal
Accident



Windscreen

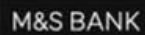


Courtesy Car

£538.26

Total Excess: £150

View details >



Legal
Assistance
+£26.29



Breakdown
Cover
+£31.54



Personal
Accident



Windscreen



Courtesy Car

£571.66

Total Excess: £150

View details >



Legal
Assistance
+£30.99



Breakdown
Cover
+£43.99



Personal
Accident



Windscreen



Courtesy Car

£573.39

Total Excess: £150

View details >



Legal
Assistance
+£30.99



Breakdown
Cover
+£43.99



Personal
Accident



Windscreen



Courtesy Car

£577.87

Total Excess: £150

View details >



Civil Service Learning

This screenshot is from 2018, taken by the ODI in response to a story regarding claims by The Sun newspaper regarding potentially 'racist' car insurance quotes by insurance firms including M&S and Admiral. To verify the story, we ran the experiment ourselves, however as The Sun had contacted these companies for quotes, they had adjusted their algorithms, while several others hadn't yet. Citizen's advice UK has since run more evaluations on discriminatory pricing, and you can read one of the latest ones [here](#).

GoCompare Car: Ford Fiesta Ghia 2002-2008 1.6 Petrol
Profession: Insurance Director
Address: Milford Haven (PPI Company :P)

Always resided in UK
Date of birth: 01/01/1980
No claims and license: 16 years
Car kept on drive

Adjust cover 37 car quotes found, 5 telematics quotes included. Sort: Annually

Hi Mohammed!

Insurance Company	Legal Assistance	Breakdown Cover	Personal Accident	Windscreen	Courtesy Car	Total Price	Total Excess	Action
LLOYDS BANK	+£30.99	+£43.99	✓	✓	✓	£1,446.32	£150	View details >
HALIFAX	+£30.99	+£43.99	✓	✓	✓	£1,458.04	£150	View details >
BANK OF SCOTLAND Decisions well made	+£30.99	+£43.99	✓	✓	✓	£1,459.64	£150	View details >
M&S BANK	✓	✓	✓	✓	✓	£1,476.70	£150	View details >



Automated Inference on Criminality using Face Images

Xiaolin Wu

McMaster University

Shanghai Jiao Tong University

xwu510@gmail.com

Xi Zhang

Shanghai Jiao Tong University

zhangxi_19930818@sjtu.edu.cn



“Unlike a human examiner/judge, a computer vision algorithm or classifier has absolutely no subjective baggages [sic], having no emotions, no biases whatsoever due to past experience, race, religion, political doctrine, gender, age, etc., no mental fatigue, no preconditioning of a bad sleep or meal. The automated inference on criminality eliminates the variable of meta-accuracy (the competence of the human judge/examiner) all together.”

—Wu & Zhang(2017)



“Unlike a human examiner/judge, computer vision algorithm or classifier has absolutely no subjective baggages [sic], no emotions, no biases whatsoever due to experience, race, religion, political affiliation, gender, age, etc., no mental fatigue, no conditioning of a bad sleep or meal. The automated inference on criminality eliminates the variable of meta-accuracy (the competence of the human judge/examiner) all together.”

Nonsense

—Wu & Zhang(2017)

Example from “calling bullshit” lecture series from the University of Washington



Good or bad ideas?

1. a tool that analyses the sentiment of a user's tweets, assesses whether they are suicidal and alerts friends	4. a risk-assessment tool that uses AI to advice on prison sentences based upon criminal profile analysis
2. automatic pricing algorithm for taxi firm which responds to surges in demand	5. using energy efficiency data and winter fuel allowance data to target efficiency advice
3. using performance data to advice on how to save money in the emergency services	6. publishing genomes of 100,000 individuals for use in public health



Data ethics

More on the examples used in this section (as well as more examples) -

Guidance

Data Science Ethical Framework

This framework is intended to give civil servants guidance on conducting data science projects, and the confidence to innovate with data.

Published 19 May 2016

Last updated 13 June 2018 — [see all updates](#)

From: [Cabinet Office](#), [Government Digital Service](#), and [The Rt Hon Matt Hancock MP](#)

This publication was withdrawn on 13 June 2018

This guidance has been replaced by the [Data Ethics Framework](#).

<https://www.gov.uk/government/publications/data-science-ethical-framework>



Data ethics canvas



<https://theodi.org/article/data-ethics-canvas/>



Course materials

<http://training.theodi.org/csl-an2/>

AN2 - Data and analysis

Welcome. This page accompanies the **AN2 - Data and analysis** course and provides access to slides exercises and related case studies.

The main aim of the course is to *equip policy makers with the knowledge and skills they need to effectively use open data in policy making.*

Key outcomes of the course include:

- Define data, open data and big data
- Describe how data is used to inform policy making
- Identify the benefits of open data to policy making
- Perform some basic statistical analysis on data to identify the shape and trends in data.
- Access the ethical risks of using data in policy making
- Perform a practical piece of policy making with real data





Session 4

Outcomes

6. List the stages in carrying out data analysis for policy making.
7. Create a plan for carrying out data analysis for policy making.
8. Review the role of open data in policy making

Self-study

9. Carry out a simple data analysis using a number of tools.
10. Create an interactive data visualisation.
11. Communicate the results of a data analysis to decision makers.



Civil Service
Learning

What has been the impact of closing 10 fire stations in London?



Data analysis practical

You have been tasked with saving money in the local fire service.

You must propose a solution that saves money while minimising the impact on service delivery.

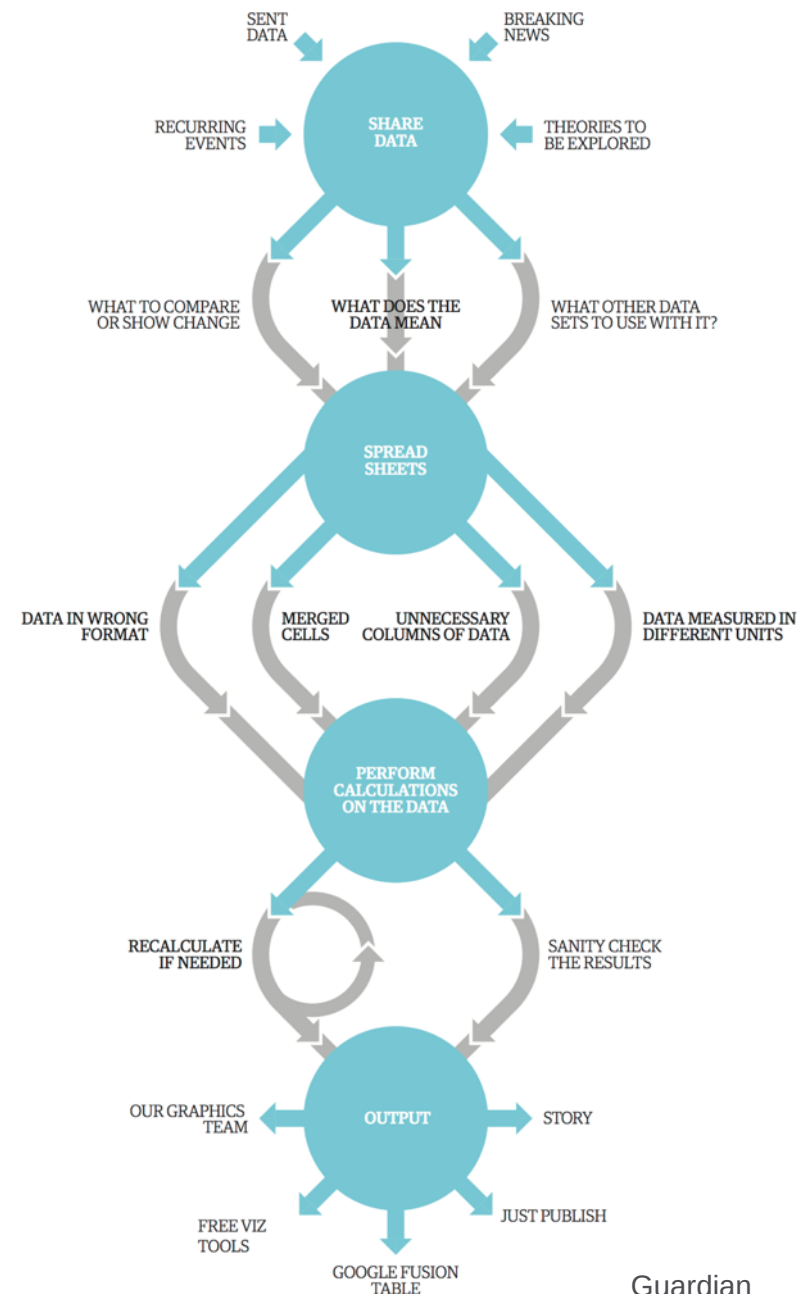
Which factor do you think is the most significant in analysing performance to identify savings (top of the decision tree)



Paul Hudson

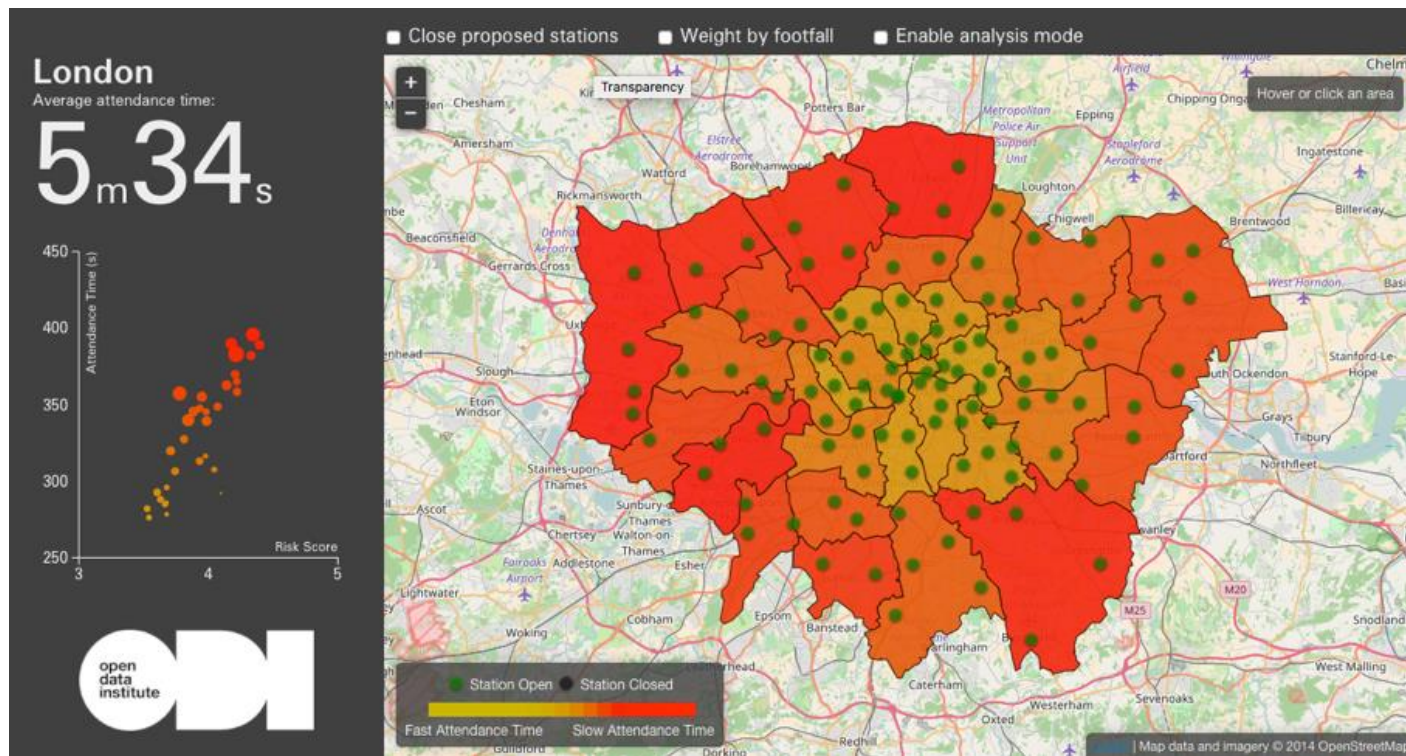
Using data in analysis

1. Collect data.
2. Establish context of data.
3. Clean the data.
4. Translate, filter and merge data.
5. Initial evaluation.
6. Go back to start?
7. Secondary evaluation and analysis.
8. Sanity check.
9. Create output.





<http://london-fire.labs.theodi.org/explore/>



<http://london-fire.labs.theodi.org/map.html>



Stage 1: collect data

Make a space on your group desk

Put a yellow post-it note in the middle and draw a fire on it. Write a time on it between 4 minutes and 7 minutes.

Scatter a series of all different colour (including yellow) post-it around this one to keep it in the middle. Write a time between 4 and 7 minutes on each one.

Stage 2: establish context

Each post-it note represents an incident that has been responded to by the London fire brigade.

Each colour represents a different fire stations from which the appliance (fire engine) was sent



Stage 3: clean the data

Given the context does your data make sense?

Stage 4: Translate, filter and merge data

Nothing to do...



Stage 5: initial evaluation

How do we calculate the impact of closing the YELLOW fire station (and specifically the middle incident)?

What is your design?

Stage 6: Go back to start?

Did it work? Is it logical?

Stage 7: Secondary data analysis

Can we apply this technique to all of the data? How?

(see self-study)



Stage 8: Sanity check

Does your design overfit (or use n th degree polynomial?)

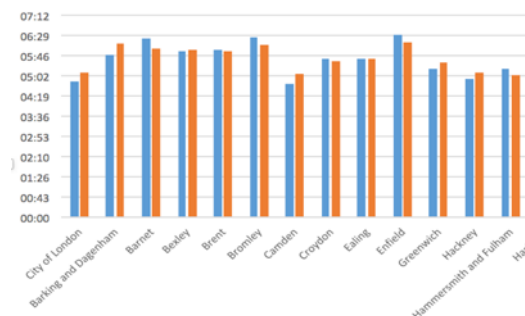
Can your design really model the future or just show how the past was?



Stage 9: Creating output

There are many options for output. Consider the best one for your purpose.

- A map can be ideal for such geographic datasets (search for London Data Store borough Excel KML).
- A chart showing a comparison of before and after statistics can be easily generated in excel.
- An highly interactive dashboard can be created with dataseedapp.com.





Fire station closures

The main exercise in the afternoon looked at analysing the impact of closing fire stations in London as proposed in the [5th London safety plan](#). The Open Data Institute built a policy tool to help people engage with the conversation and explore the impact themselves. Explore the tool below and follow the links to have a look at the impact yourself in the related exercises.

[Explore the ODI policy tool](#)

[Download the Excel exercise](#)

[Download the same exercise for Tableau](#)

[Download the same exercise for RStudio](#)

The exercises also contain the links to the [datasets](#) and excel [mapping file](#) for London.





Review

London Fire Stations analysis was a story started by a participant in an ODI Training course like this one. What is your story? Who and what would you need to help you?

If I'm a policy maker, what do I now need to go and do?

What do I need to go and do next?

What is your main takeaway from the course overall?

What are you going to apply back in the workplace?



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Thank you

We hope you enjoyed this
experience brought to you by

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Now over to you! What are you going to do differently?